Anatoly Zavyalov

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Research Interests

Differential privacy, property testing, graph algorithms, sublinear algorithms

Education

Boston University — Sep 2024 - Jun 2029 (expected)
Ph.D. Student in Computer Science — Advised by Sofya Raskhodnikova

University of Toronto — Sep 2020 - Jun 2024
H.B.Sc. in Applied Mathematics — 3.95 cGPA

Selected Publications

1. S. Raskhodnikova, A. D. Smith, C. Wagaman, **A. Zavyalov**, "Local Node Differential Privacy". arXiv preprint arXiv:2602.15802 (2026).
2. J. Shallit, **A. Zavyalov**. "Transduction of Automatic Sequences and Applications". *CIAA 2023*. Lecture Notes in Computer Science, vol 14151.
3. A. D. Hincks, **A. Zavyalov**, D. Bansal. "A graph database solution for tracking the deployment and layout of a large radio interferometer". *SPIE Astronomical Telescopes + Instrumentation 2022*. Proc. SPIE 12189.

Honors and Awards

Chair's Graduate Fellowship - $10,000 USD — Feb 2025
Awarded on admission to Boston University.

NSERC Undergraduate Student Research Award - $7,500 CAD — Mar 2023
Awarded by U of T Department of Computer Science.

SURP Fellowship & U of T Excellence Award - $17,095 CAD — Apr-May 2021
Combined funding for a research fellowship at the Dunlap Institute of Astronomy.

NSERC USRA ($7,500 CAD) & Fields USRP — Declined — Mar 2022
Received and declined offers for two separate summer research programs.

University of Toronto Scholar - $3,000 CAD — Aug 2021, Aug 2022
Awarded by the University of Toronto for high academic achievement ($\$1,500 \times 2$).

Dean's List Scholar — 2021, 2022, 2023
Awarded for a cumulative GPA of 3.50 or higher.

Trinity College Academic Achievement Scholarships - $1,500 CAD — Nov 2021 - Nov 2023
Awarded by Trinity College for high academic achievement ($\$500 \times 3$).

Talks

1. **"Transduction of Automatic Sequences and Applications"**, CIAA 2023. (30 min) **(Slides)**
2. **"Automatic Sequences"**, Canadian Undergraduate Math Conference, 2023 (30 min) **(Slides)**
3. **"Automatic Sequences"**, UTSC CMS Undergraduate Seminar, 2023 (60 min) **(Slides)** **(Recording)**

Research Experience

Boston University | PhD Student — Sep 2024 - Present
Department of Computer Science

- Working in the areas of graph differential privacy and graph property testing, supported by a **Chair's Graduate Fellowship**.

University of Waterloo | Research Assistant — May 2022 – Jul 2022, May 2023 – Jul 2023
David R. Cheriton School of Computer Science

- Researched and implemented algorithms into **Walnut**, a theorem proving software for automatic sequences written in **Java**, working with Professor Jeffrey Shallit.
- Research culminated in a publication and presentation at *CIAA 2023*.

University of Toronto | Research Assistant — May 2023 – Aug 2023
Department of Computer Science

- Researched algebraic methods for concurrent program verification and race condition detection, supported by an **NSERC Undergraduate Student Research Award (NSERC USRA)**.

University of Toronto | Research Fellow — May 2021 – Apr 2022
David A. Dunlap Department of Astronomy and Astrophysics

- Developed **Padloper**, a full-stack graph database solution for tracking deployment and layout of a large radio interferometer, using **JanusGraph, Flask** and **React**, working with Professor Adam Hincks.
- Padloper will be used for the **Hydrogen Intensity and Real-time Analysis eXperiment (HIRAX)**, the **Simons Observatory**, and **Canadian Hydrogen Intensity Mapping Experiment (CHIME)**.
- Research culminated in proceedings at *SPIE Astronomical Telescopes + Instrumentation 2022*.

Teaching Experience

Boston University | Teaching Fellow

- Teaching Fellow for **CSC237: Probability in Computing** — Sep 2025 - Dec 2025

University of Toronto | Teaching Assistant

- TA for **CSC363: Computational Complexity and Computability** — Jan 2024 - Apr 2024
- Lead TA for **CSC373: Algorithm Design and Analysis** — Sep 2023 - Dec 2023
- TA for **CSC240: Enriched Introduction to the Theory of Computation** — Feb 2022 - May 2022

STEM Outreach

SigmaCamp | Academic Administrator & Counselor — 2022 - Present

- Directing *Problem of the Month*, a free international STEM competition for teenagers, which received over 2,000 solution submissions from 130+ students in 2024-2025.
- Assembled and managed a team of 12 subject leads and dozens of graders. Also managed the Qualification Quiz entrance exam, which received 650+ solutions submissions in 2025.

Skills

- **Human Languages**: English (C2), Russian (C2), German (B1), Portuguese (A2)
- **Programming Languages**: Python, JavaScript, C++, Java, PostgreSQL, LaTeX
- **Frameworks/Libraries**: ReactJS, Flask, NumPy, PixiJS